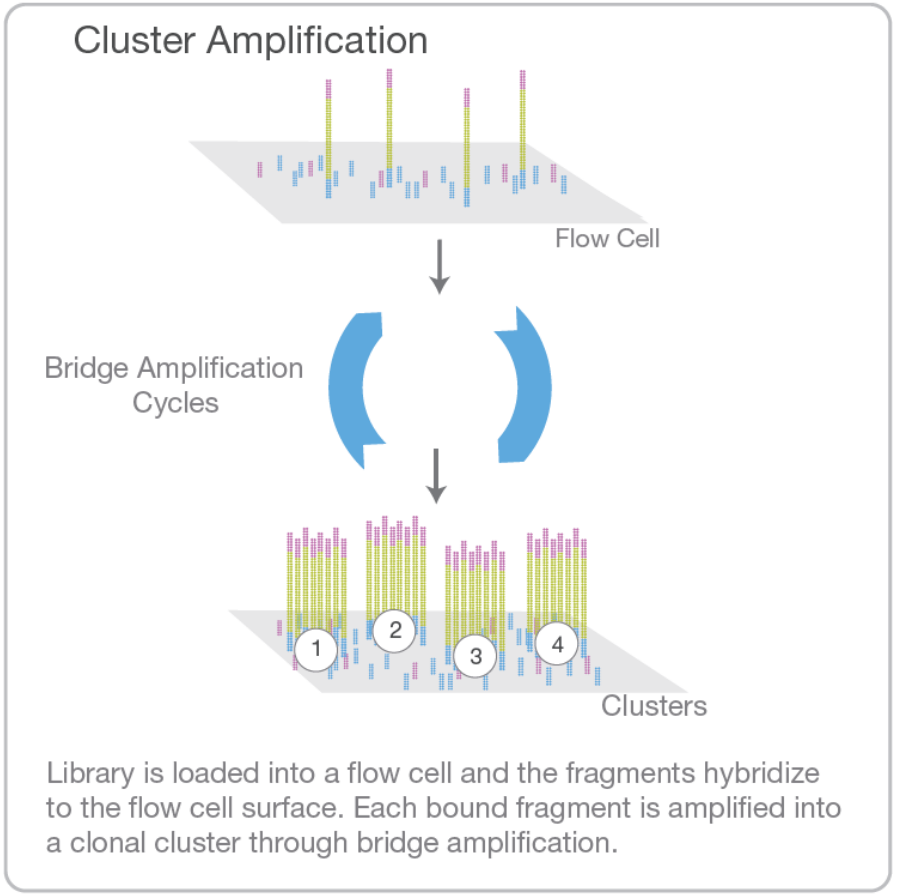
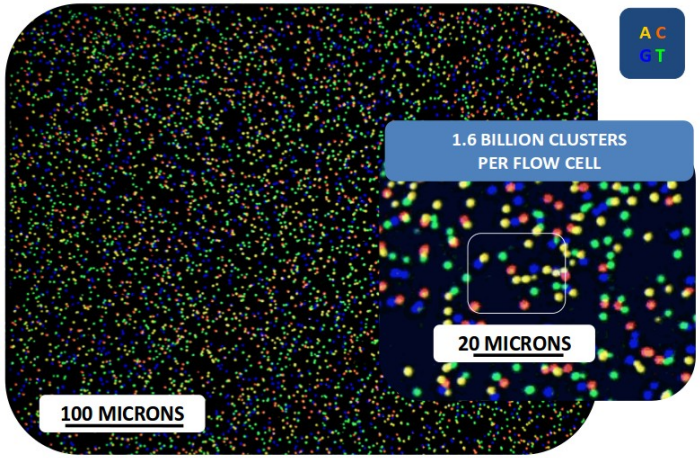
Library prep QC

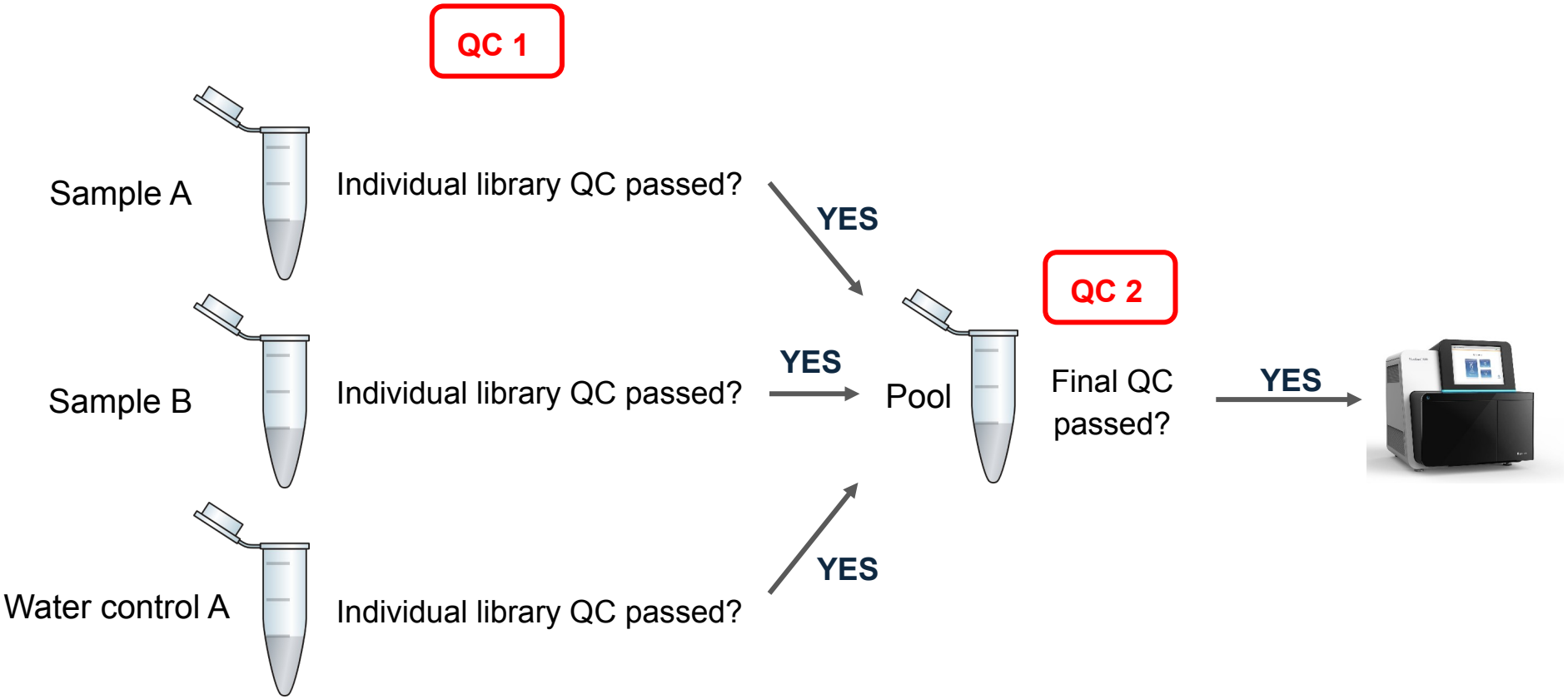
Cluster amplification is essential for Illumina sequencing



Optimal cluster formation is dependent on:
~ loading concentration
~ fragment length

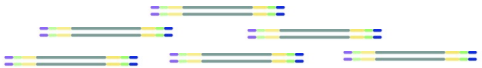


Library prep QC

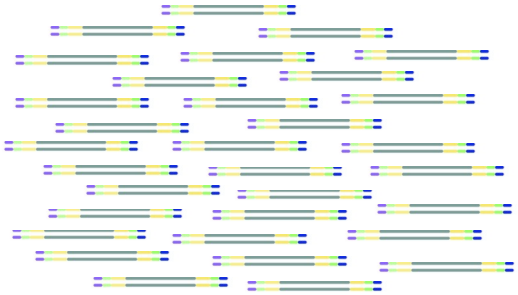


Library prep QC

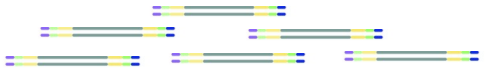
Concentration



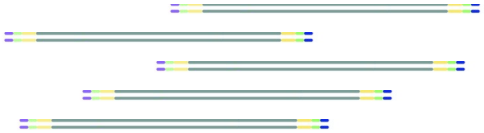
vs



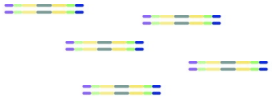
Length



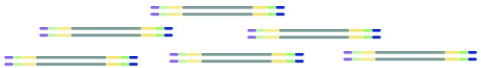
vs



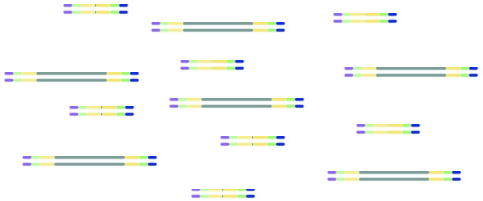
or



Remove adapter dimers



vs



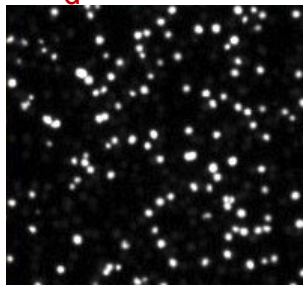
Library prep QC - Concentration

Why is an accurate concentration important?

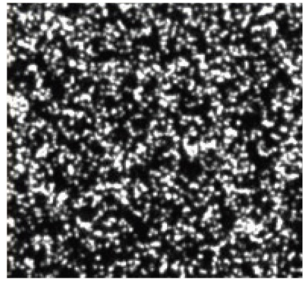
- Individual samples: to pool equal amounts of different samples together
  - Aim for an even amount of sequencing data per sample

- Final library: sequencer requires loading with a very precise concentration
  - If concentration is overestimated (actual conc. is lower than what you think)
    - Underloading and underclustering of sequencing run
    - Sequencing run will yield less sequencing data than usual
  - If concentration is underestimated (actual conc. is higher than what you think)
    - Overloading and overclustering of sequencing run
    - Sequencing run will suffer from higher error rates and reduced data output

Underloaded



Overloaded



Library prep QC - Concentration

DO NOT USE to quantify libraries

Spectrophotometry
(e.g. Nanodrop)



- DNA absorbs light most strongly at 260 nm
- Level of absorbance ~ concentration of DNA

BUT other molecules (RNA, proteins) & contaminants also absorb light at 260 nm, causing up to 10-fold **overestimation** of the DNA concentration!

Fluorometry
(e.g. Qubit, qPCR)





- Detects fluorescent dyes that only bind DNA
- Most reliable method for NGS

Electrophoresis
(e.g. Tapestation, Bioanalyzer)



- Compare intensity of DNA bands to standards
- Especially useful to determine concentration in certain fragment length range

Library prep QC - Concentration

Qubit dsDNA High Sensitivity (HS) Assay kit



- dsDNA HS Reagent (200x concentrate)
- dsDNA HS Buffer
- dsDNA HS Standard #1
- dsDNA HS Standard #2

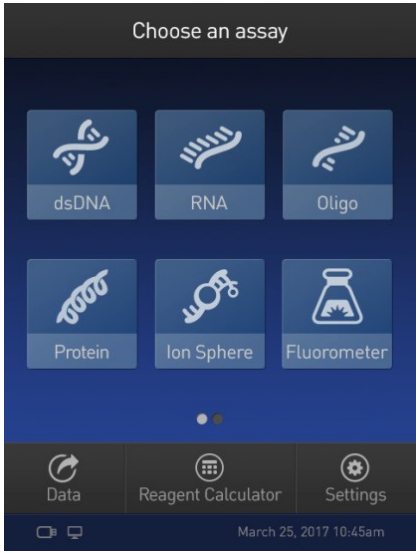
1. Prepare Working Solution:
   - Diluting HS reagent 1:200 in HS buffer

1. Dilute in Working Solution:
   - Standards: 10 µl standard + 190 µl Working Solution
   - Samples: 1 µl sample + 199 µl Working Solution

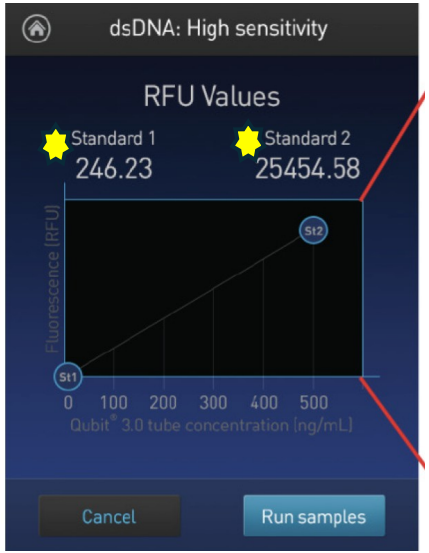
1. Vortex 2-3 seconds before running standards/samples

Library prep QC - Concentration

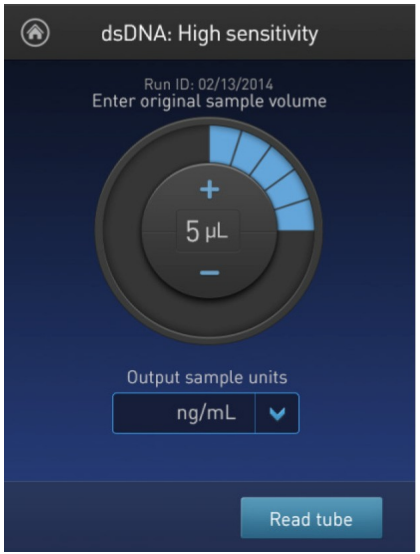
Qubit dsDNA High Sensitivity (HS) Assay kit



Select dsDNA HS workflow



Run the 2 standards



Enter sample volume added to tube



Run sample

Top value = original conc.

Bottom value = diluted conc.

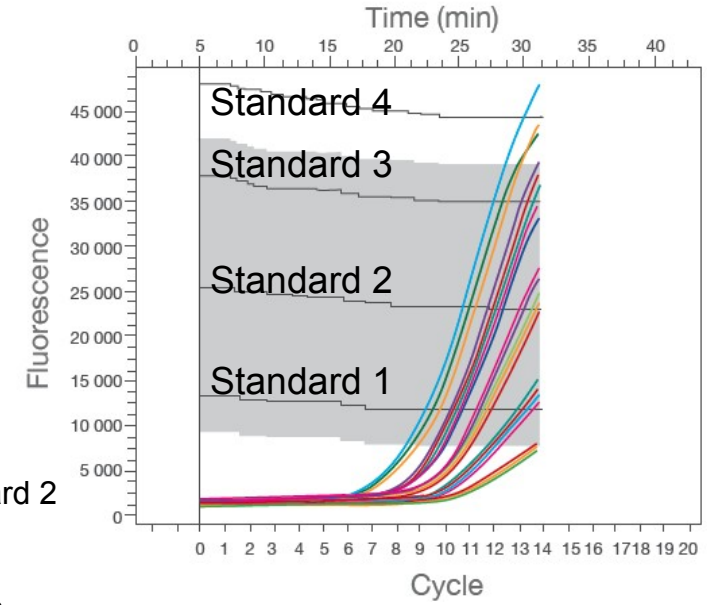
We recommend keeping a log of the RFU values every time the standards are measured.
Over time the fluorescence decays and can inadvertently give erroneous quantifications

Library prep QC - Concentration

- **What if the concentration is too low?**
  - In future runs, consider adding more cycles to the indexing PCR step
  - Amplify library using P5/P7 primers and qPCR-based library amplification kit

1. Set up qPCR reaction: library + mastermix + P5/P7 primers
2. Treat standards as separate samples (do not add mastermix)
3. Start run

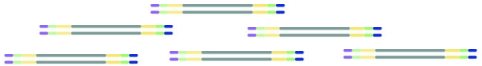
Temperature	Time	Cycles
98°C	45 sec	1
98°C	15 sec	25
63°C	30 sec	
72°C	1m 45 sec	
Plate read		
72°C	20 sec	



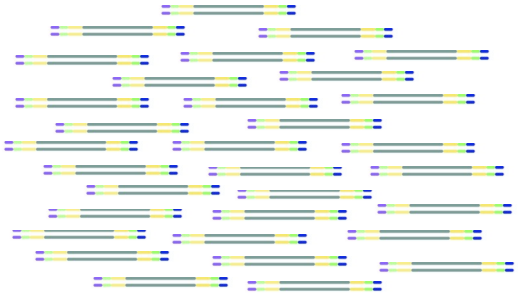
1. Pause cycler during the 20s at 72°C step when a sample crossed the Standard 2 line, remove the sample and continue amplification for other samples
2. After amplification, perform bead cleanup on amplified libraries and QC again

Library prep QC

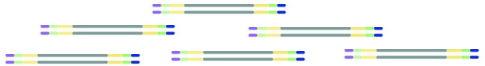
Concentration



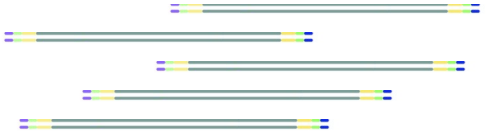
vs



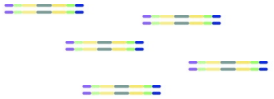
Length



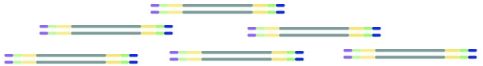
vs



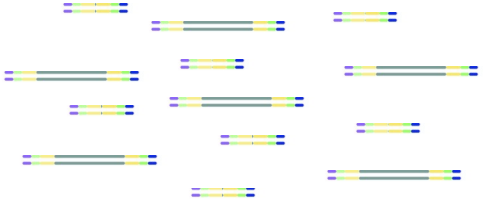
or



Remove adapter dimers



vs



Library prep QC - Length

Why is it important to assess fragment length in a library?

- Long fragments (>1.2 kb) do not cluster on Illumina flow cells and are therefore not sequenced
- Short fragments are preferentially amplified (better clustering) and will thus be preferentially sequenced
- Fragment length distribution impacts calculations to go from ng/µl (Qubit) to nM for loading concentrations

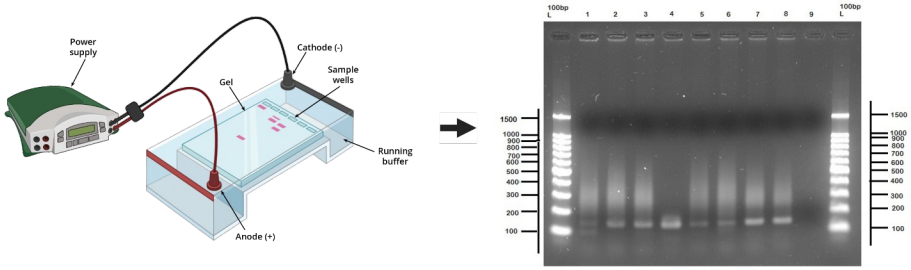
Qubit

$$\text{Concentration in nM} = \frac{\text{concentration in ng/µl}}{\text{660g/mol x average library size in bp}} \times 10^6$$

Capillary or gel electrophoresis

Library prep QC - Length

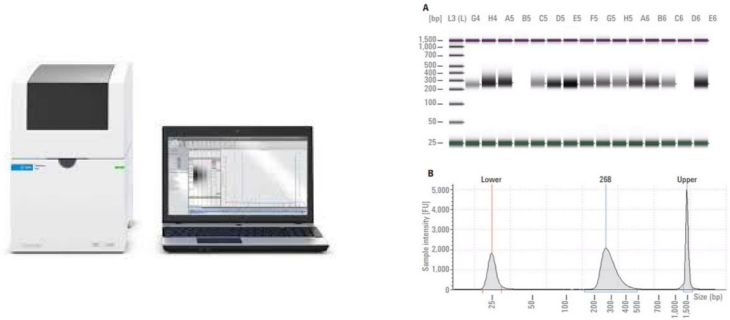
Gel electrophoresis



1. PCR amplify an aliquot of the library using universal P5/P7 Illumina primers
2. Gel electrophoresis of ladder and samples
    - Takes ~ 4 hrs

Capillary electrophoresis

E.g. Tapestation, Bioanalyzer



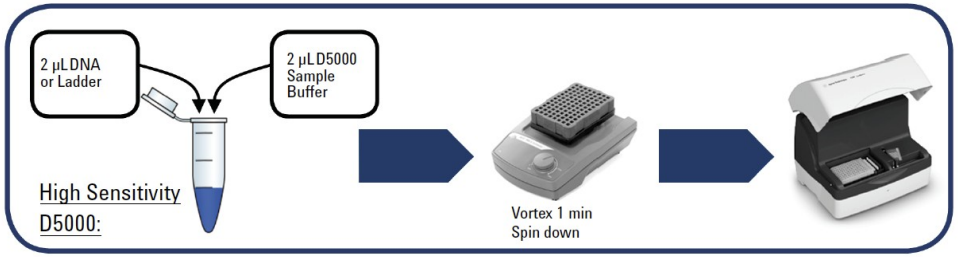
1. Add buffer to aliquot of library
2. Run machine

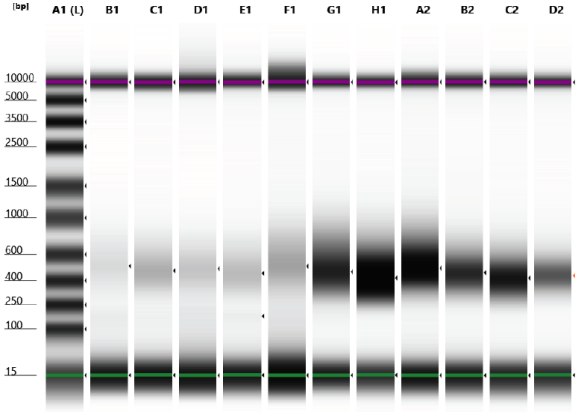
- Tapestation ~ 10 minutes (depending on # samples)
- Bioanalyzer ~ 45 minutes
- No PCR bias
- Can calculate concentration in nM

Library prep QC - Length

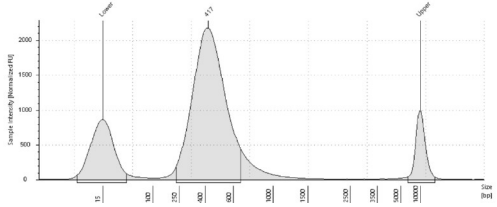
Tapestation

- High Sensitivity D1000 or D5000 kits (1kb or 5kb detection limit)





Peak: H1

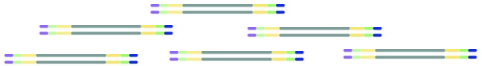


Location	H1
Concentration	1470
Description	
Observations	

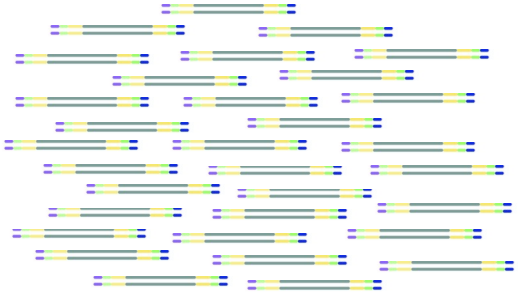
Size [bp]	Calibrated Conc. [pg/µl]	Assigned Conc. [pg/µl]	PeakMolarity [pmol/l]	% Integrated Area	PeakComment	Observations
15	346	-	35500	-		Lower Marker
417	1470	-	5430	100.00		
10000	180	180	27.7	-		Upper Marker

Library prep QC

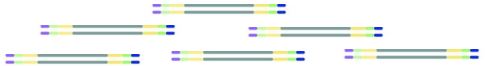
Concentration



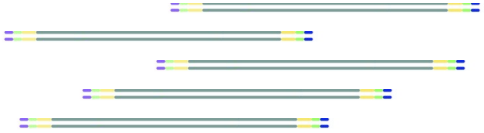
vs



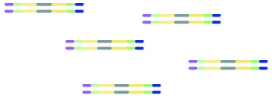
Length



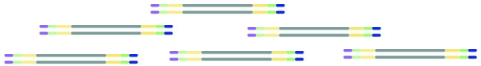
vs



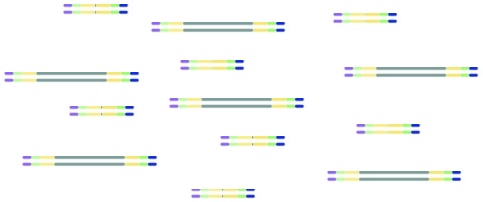
or



Remove adapter dimers

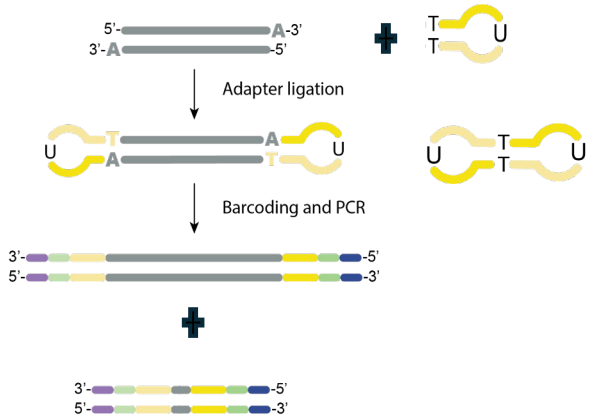


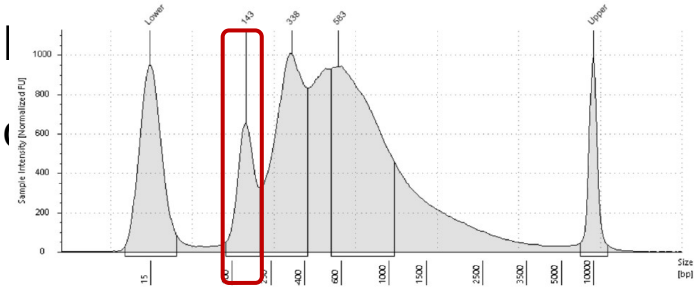
vs



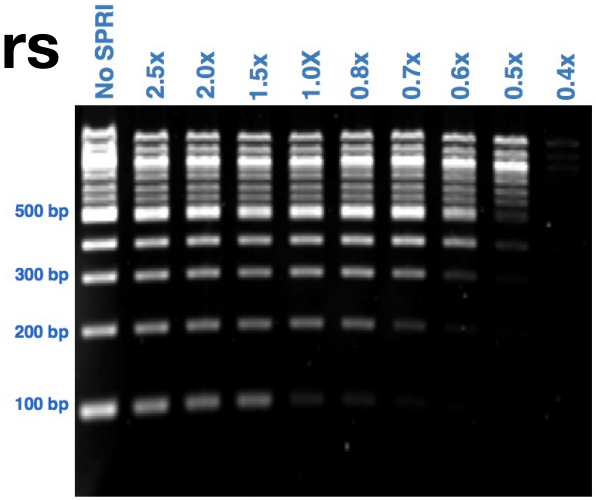
Library prep QC - Clean up adapter dimers

- Adapter dimers are:
  - A byproduct of the adapter ligation step during library prep (TruSeq/NEBNext Ultra II DNA kits)
  - ± 140 bp long at the final QC step (after barcoding PCR)
  - Use capillary electrophoresis (preferred) or





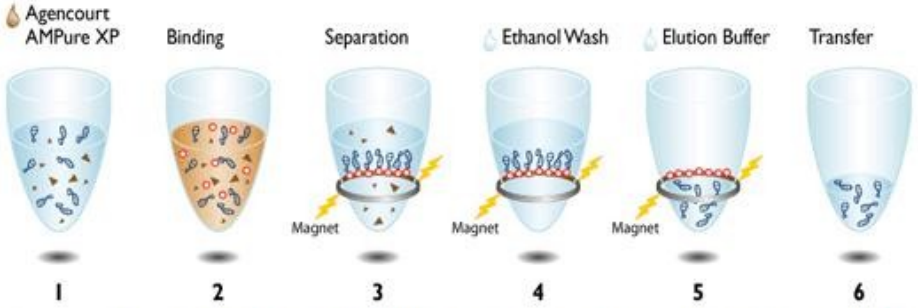
Library prep QC - Clean up adapter dimers



AMPure XP SPRI select beads (paramagnetic)

or homemade SPRI beads (>10x cheaper)

- The longer a fragment, the better it will bind to the beads
- The fewer beads you add, the more short fragments you'll lose
- **Perform size selection using these beads to remove adapter dimers**



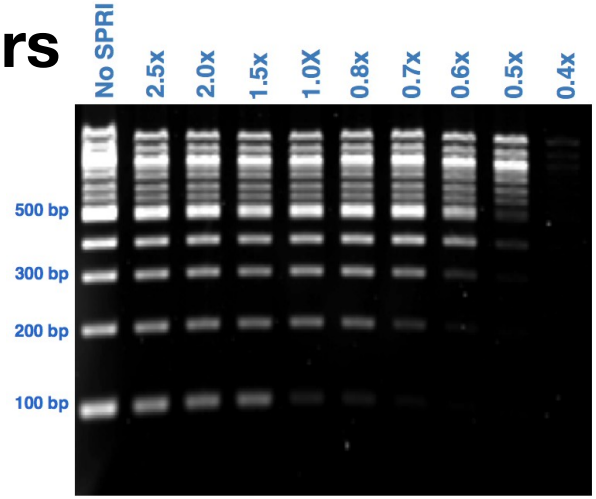
1. PCR reaction 2. Binding of PCR amplicons to magnetic beads 3. Separation of PCR amplicons bound to magnetic beads from contaminants 4. Washing of PCR amplicons with Ethanol 5. Elution of PCR amplicons from the magnetic particles 6. Transfer away from the beads into a new plate

$$\text{SPRI ratio} = \frac{\text{Volume beads}}{\text{Volume reaction}}$$

Example: for a 1.8x SPRI clean-up:

- Add 18 µl of SPRI beads to 10 µl of sample

Library prep QC - Clean up adapter dimers

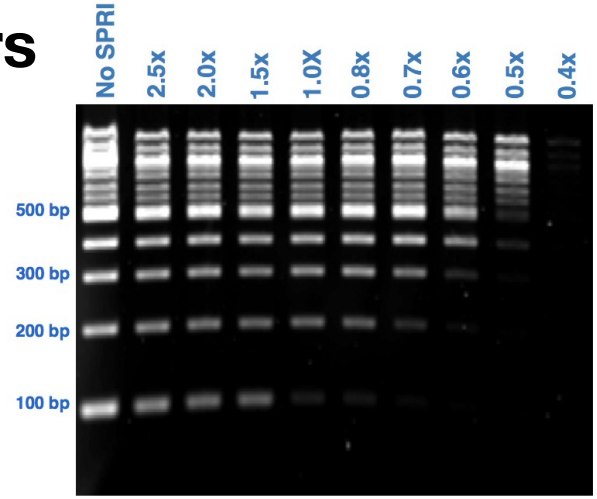


AMPure XP SPRI select beads (paramagnetic)

- Size selection using beads to remove adapter dimers
- Rule of thumb: if you see a clear trace of adapter dimers, performs another round of cleanup

Step	To remove	SPRI ratio
After adapter ligation	Adapter dimers (64 bp) & reagents	0.9x
After P7/P5 PCR	P5-adapter dimer-P7 (143 bp) & reagents	0.8x
QC of library	P5-Adapter dimer-P7 (143 bp)	0.8x

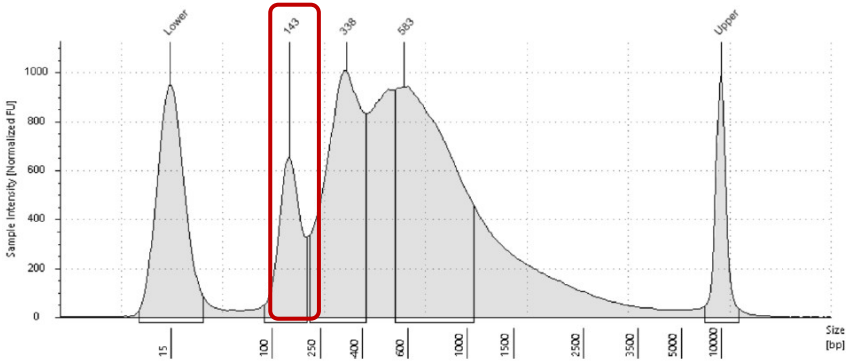
Library prep QC - Clean up adapter dimers



AMPure XP SPRI select beads (paramagnetic)

- Size selection using SPRI beads to remove adapter dimers
- Rule of thumb: if you see a clear trace of adapter dimers, performs another round of cleanup

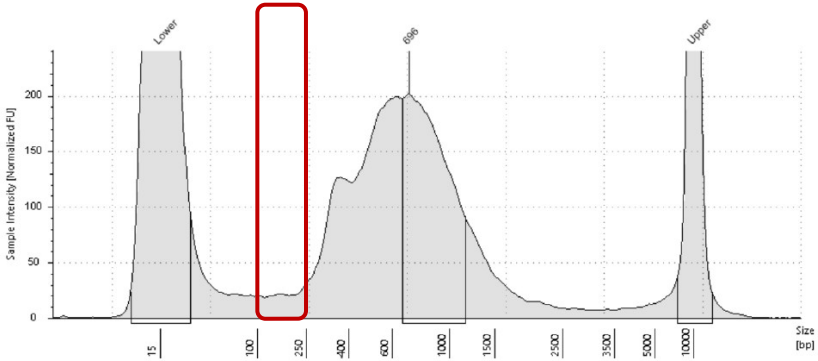
After PCR amplification



Adapter dimers

3 rounds SPRI (0.85x)

After SPRI bead clean-up

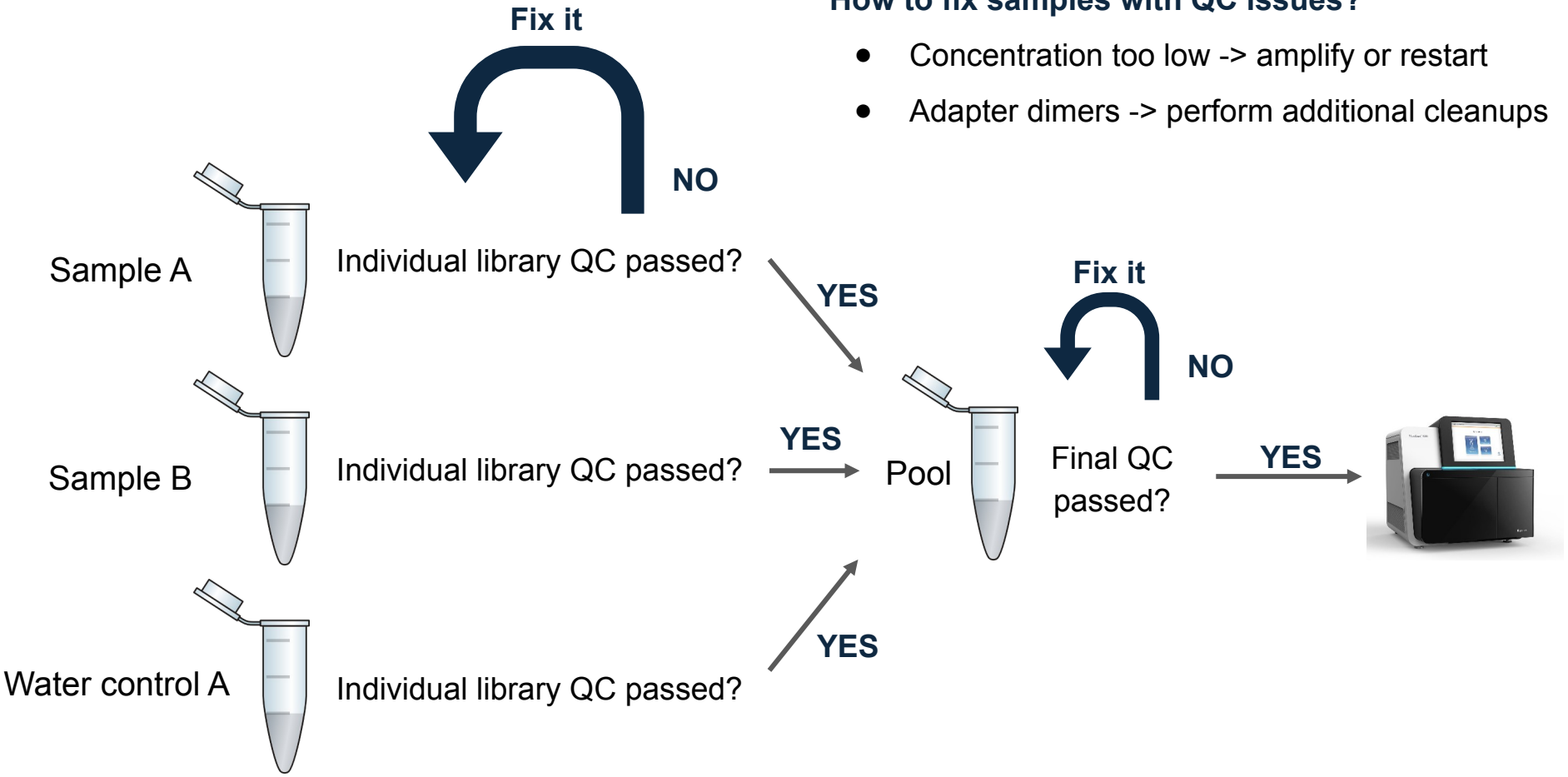


ALL DIMERS SHOULD BE REMOVED!

Library prep QC

How to fix samples with QC issues?

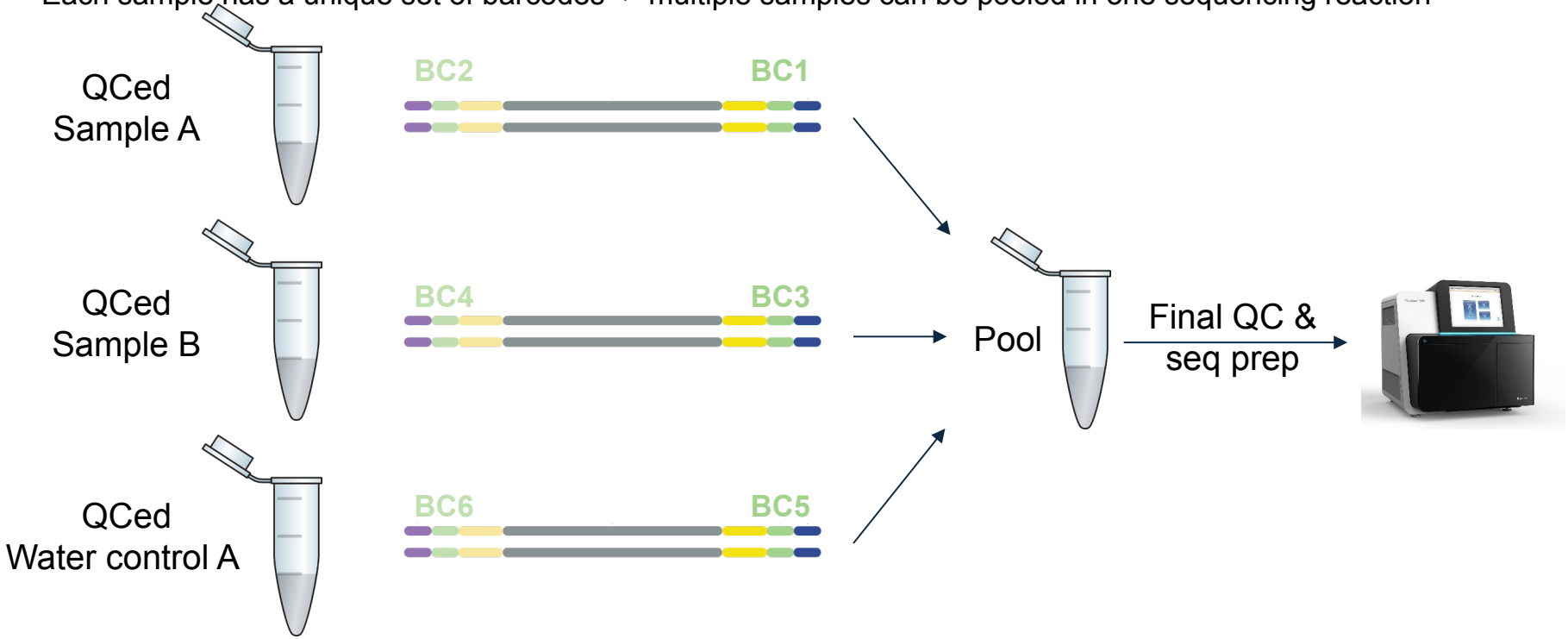
- Concentration too low -> amplify or restart
- Adapter dimers -> perform additional cleanups



Library prep QC

Pooling samples for sequencing

- Each sample has a unique set of barcodes -> multiple samples can be pooled in one sequencing reaction



Library prep QC

Pooling samples together for sequencing

1. Qubit to determine concentration of individual libraries (expect library to be 0.5 to 10 ng/ul)
2. Pool barcoded libraries so that all are equally represented (except H_2O controls, only 1:5)



	Qubit concentration		Pool		
Sample 1:	10 ng/ul	->	4.0 µl	->	40 ng
Sample 2:	5 ng/ul	->	8.0 µl	->	40 ng
Sample 3:	8 ng/ul	->	5.0 µl	->	40 ng
Sample 4:	15 ng/ul	->	2.6 µl	->	40 ng
H_2O control:	1 ng/ul	->	8.0 µl	->	8 ng
		Total:	27.6 µl		168 ng

End goal is >10 µl of 2 nM pooled library (or 4 nM for MiSeq)

Pooling guidelines

- Do not pool more than half of your library volume (in case you need to go back)
- Pool waters/negative controls less mass than samples (1/5th or 1/10th)

Library prep QC

Pick one way to calculate nM and stick with it!

- Qubit concentration and library size

  or

- Automatic calculation by Tapestation

Pooling samples together for sequencing

3. After pooling, determine quantity and quality of pool

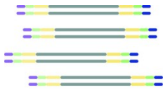
Qubit

$$\text{Concentration in nM} = \frac{\text{concentration in ng/µl}}{\text{660g/mol x average library size in bp}} \times 10^6$$

Capillary or gel electrophoresis

Example: if concentration is 6.1 ng/µl, and average library size is 550 bp -> concentration is 16.8 nM

End goal is >10 µl of 2 nM pooled library



Library prep - preparing to load the sequencer

Dilute library to the correct loading concentration:

a. iSeq:
    i. dilute to 75 pM (or another optimized concentration)
    ii. Spike in 1-5% of PhiX at the same concentration

a. MiSeq:
    i. First denature library with sodium hydroxide (NaOH)
    ii. Then dilute to 10-12 pM (or another optimized concentration)
    iii. Spike in 1-5% of denatured PhiX to denatured and diluted library

See slide decks for iSeq or MiSeq loading for more details!

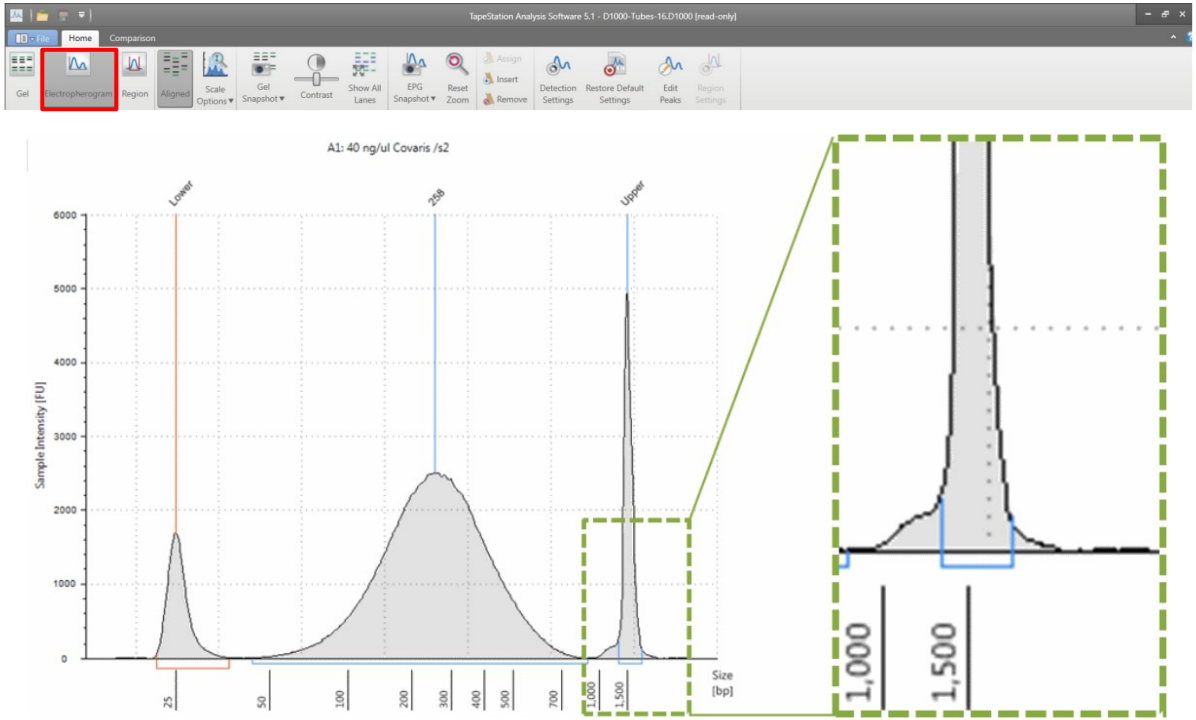
Using the Tapestation

Choosing the right assay according to your starting material

ScreenTape Assay	Quick Guide Number	
Agilent D1000 ScreenTape Assay	G2991-90031	
Agilent High Sensitivity D1000 ScreenTape Assay	G2991-90131	
Agilent RNA ScreenTape Assay	G2991-90021	
Agilent High Sensitivity RNA ScreenTape Assay	G2991-90121	RNA 500 – 10000 pg/µL
Agilent Genomic DNA ScreenTape Assay	G2991-90041	DNA 200 to >60000 bp, 10 – 100 ng/µL
Agilent D5000 ScreenTape Assay	G2991-90051	
Agilent High Sensitivity D5000 ScreenTape Assay	G2991-90151	Sequencing libraries 100 - 5,000 bp, 10 - 1,000 pg/µL
Agilent Cell-free DNA ScreenTape Assay	G2991-90060	

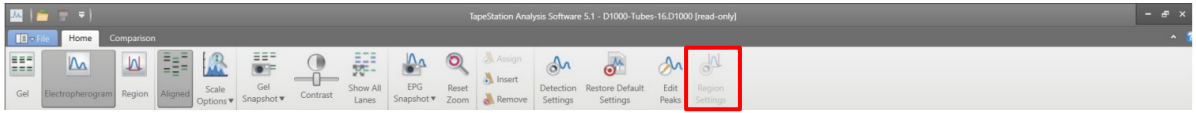
Note, sensitivity is guided by input concentration.

How to analyse a TapeStation run?

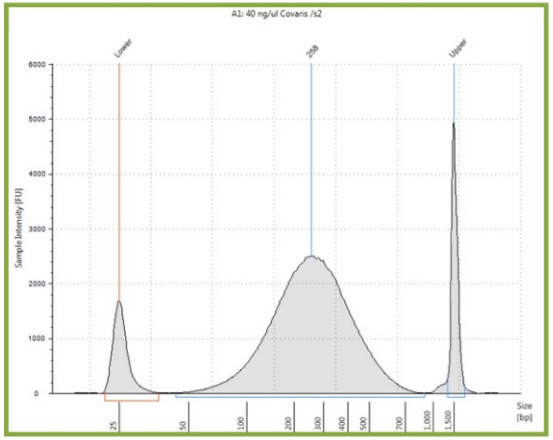


1. Ensure that all marker peaks (‘Lower’ and ‘Upper’) are properly assigned. **Concentration values are calculated using the area of the sample peak compared to the known concentration of the upper marker.**

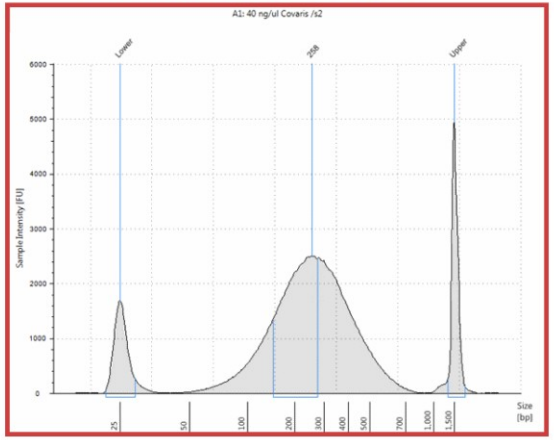
How to analyse a TapeStation run?



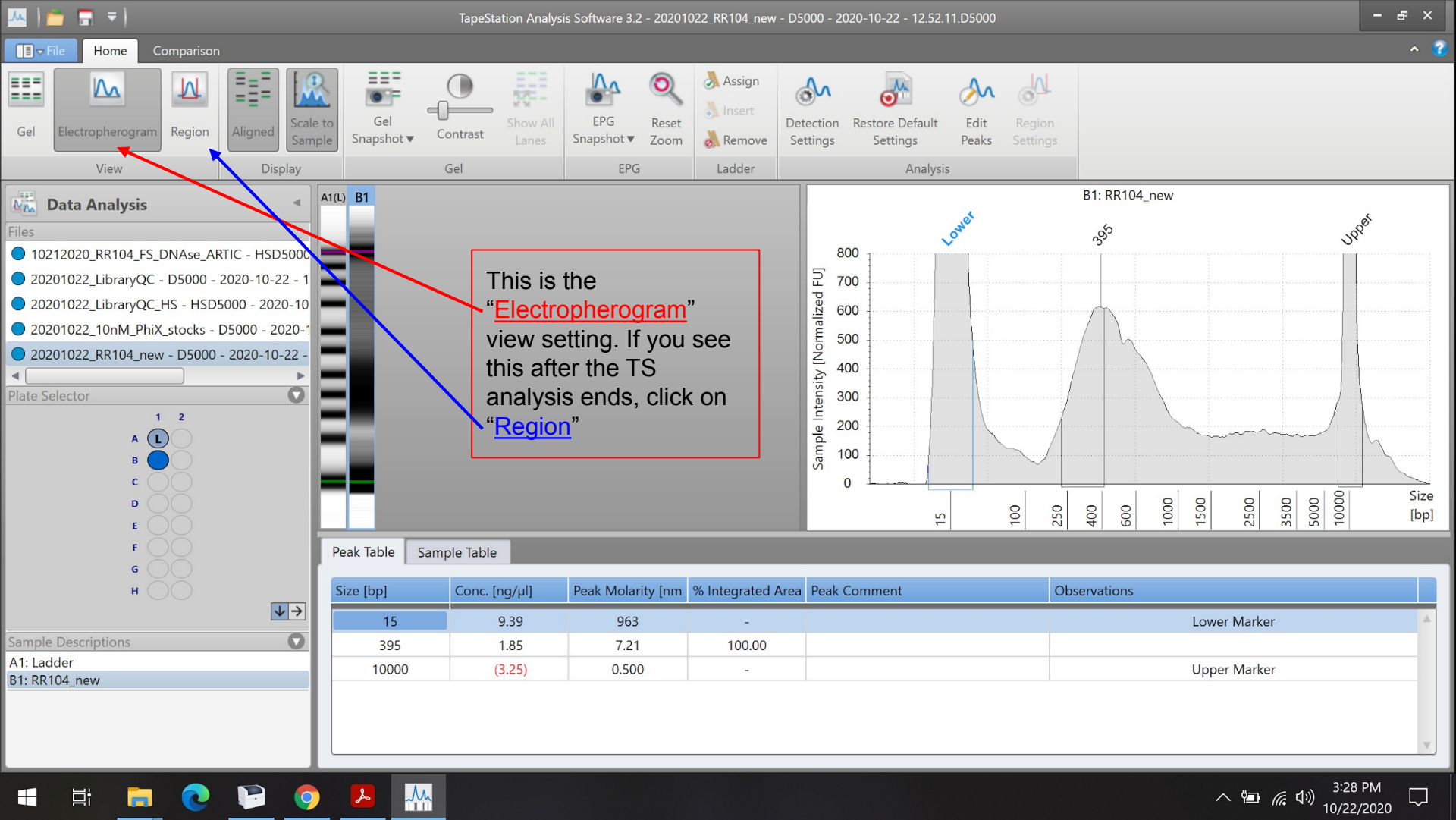
Correct peak integration
Concentration: 41.2 ng/µL



Incorrect peak integration
Concentration: 14.9 ng/µL



1. Ensure that all marker peaks (‘Lower’ and ‘Upper’) are properly assigned. **Concentration values are calculated using the area of the sample peak compared to the known concentration of the upper marker.**
2. Ensure that sample peaks are properly integrated by manually adjusting the peak when necessary.


TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000
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Edit Peaks
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EPG
Ladder
Analysis
Data Analysis
Files
10212020_RR104_FS_DNAse_ARTIC - HSD5000
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20201022_LibraryQC_HS - HSD5000 - 2020-10
20201022_10nM_PhiX_stocks - D5000 - 2020-1
20201022_RR104_new - D5000 - 2020-10-22 -
Plate Selector
Sample Descriptions
A1: Ladder
B1: RR104_new
A1(L)
B1
This is the "Electropherogram" view setting. If you see this after the TS analysis ends, click on "Region"
B1: RR104_new
Lower
395
Upper
Sample Intensity [Normalized FU]
Size [bp]
Peak Table
Sample Table
Size [bp]
Conc. [ng/µl]
Peak Molarity [nm
% Integrated Area
Peak Comment
Observations
15
9.39
963
-
Lower Marker
395
1.85
7.21
100.00
10000
(3.25)
0.500
-
Upper Marker
3:28 PM
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TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000

File | Home | Comparison

Gel | Electropherogram | Region | Aligned | Scale to Sample | Gel Snapshot | Contrast | Show All Lanes | EPG Snapshot | Reset Zoom | Assign | Insert | Remove | Detection Settings | Restore Default Settings | Edit Regions | Region Settings

View | Display | Gel | EPG | Ladder | Analysis

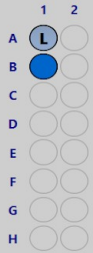
Click on “Region Settings”

Data Analysis

Files

- 10212020_RR104_FS_DNAse_ARTIC - HSD5000
- 20201022_LibraryQC - D5000 - 2020-10-22 - 1
- 20201022_LibraryQC_HS - HSD5000 - 2020-10
- 20201022_10nM_PhiX_stocks - D5000 - 2020-1
- 20201022_RR104_new - D5000 - 2020-10-22 -

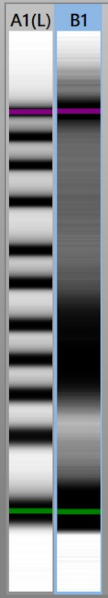
Plate Selector

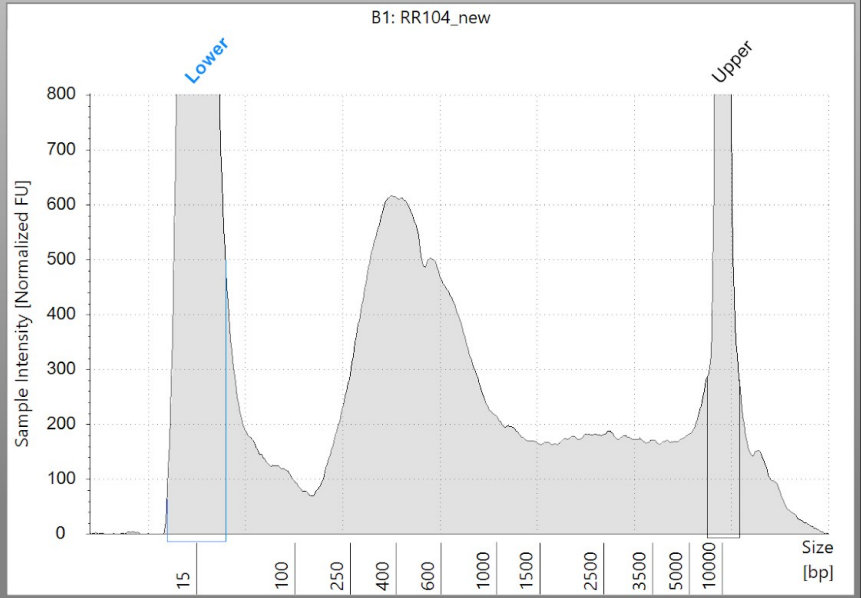


Sample Descriptions

A1: Ladder

B1: RR104_new

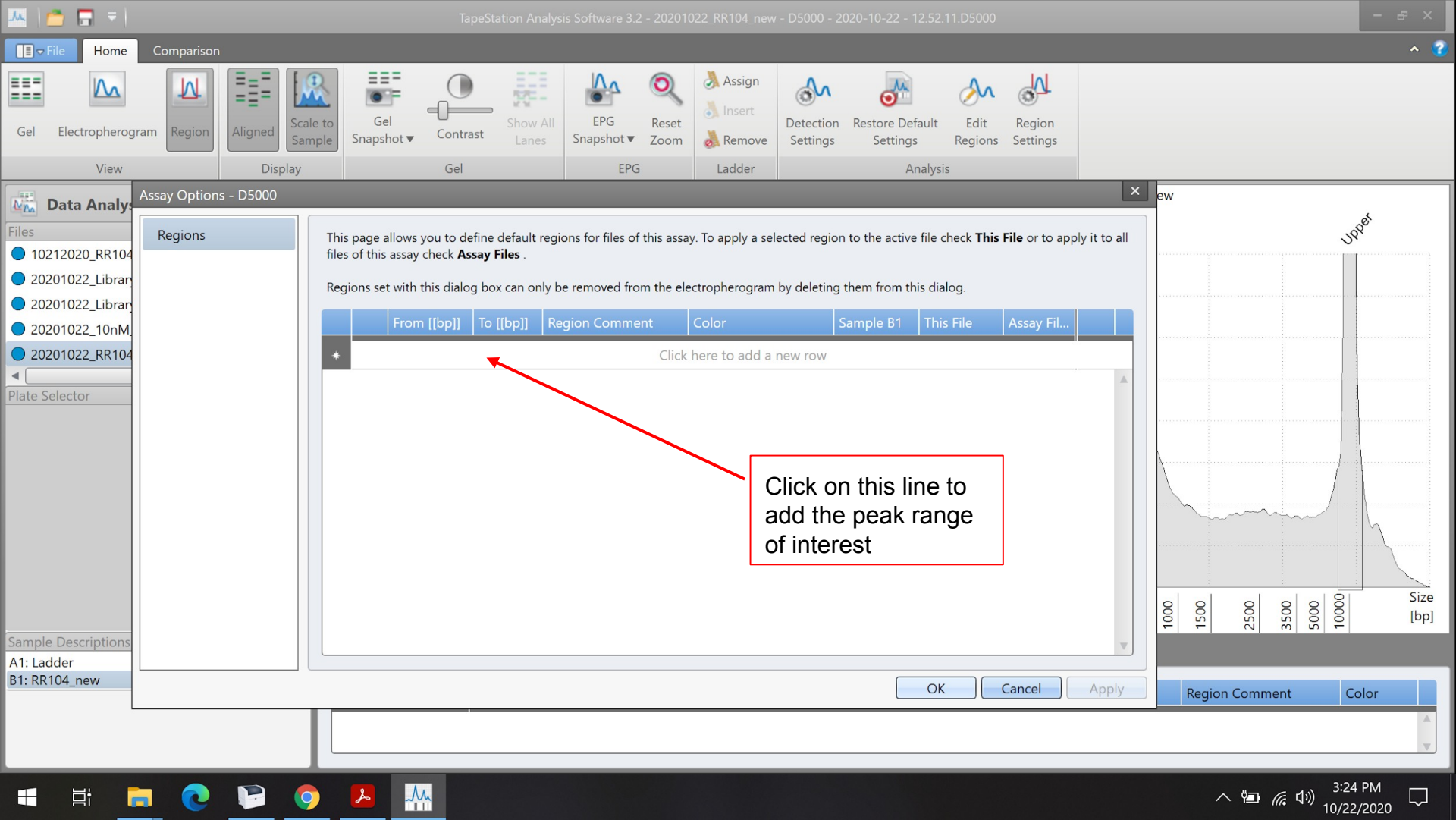


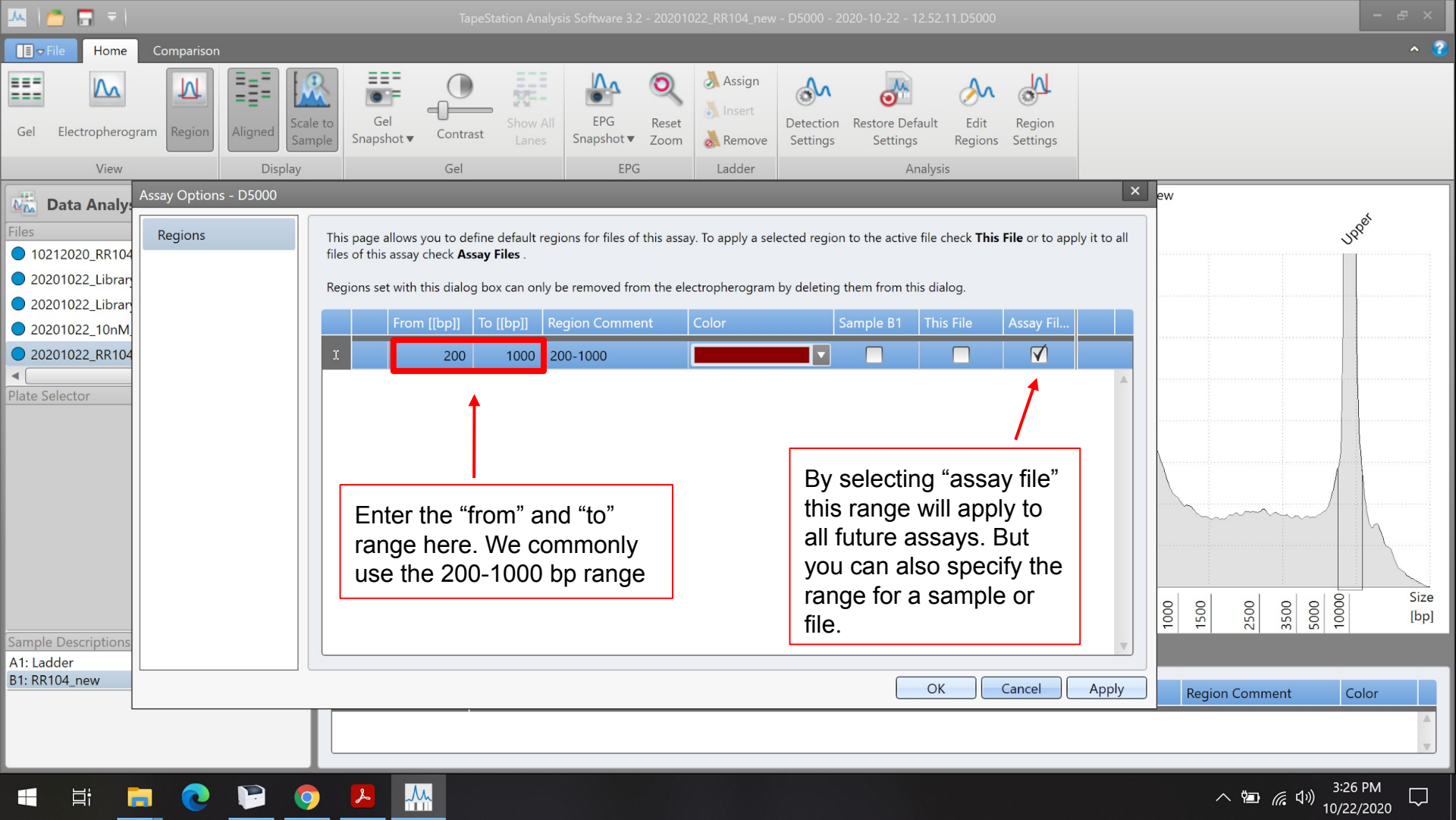


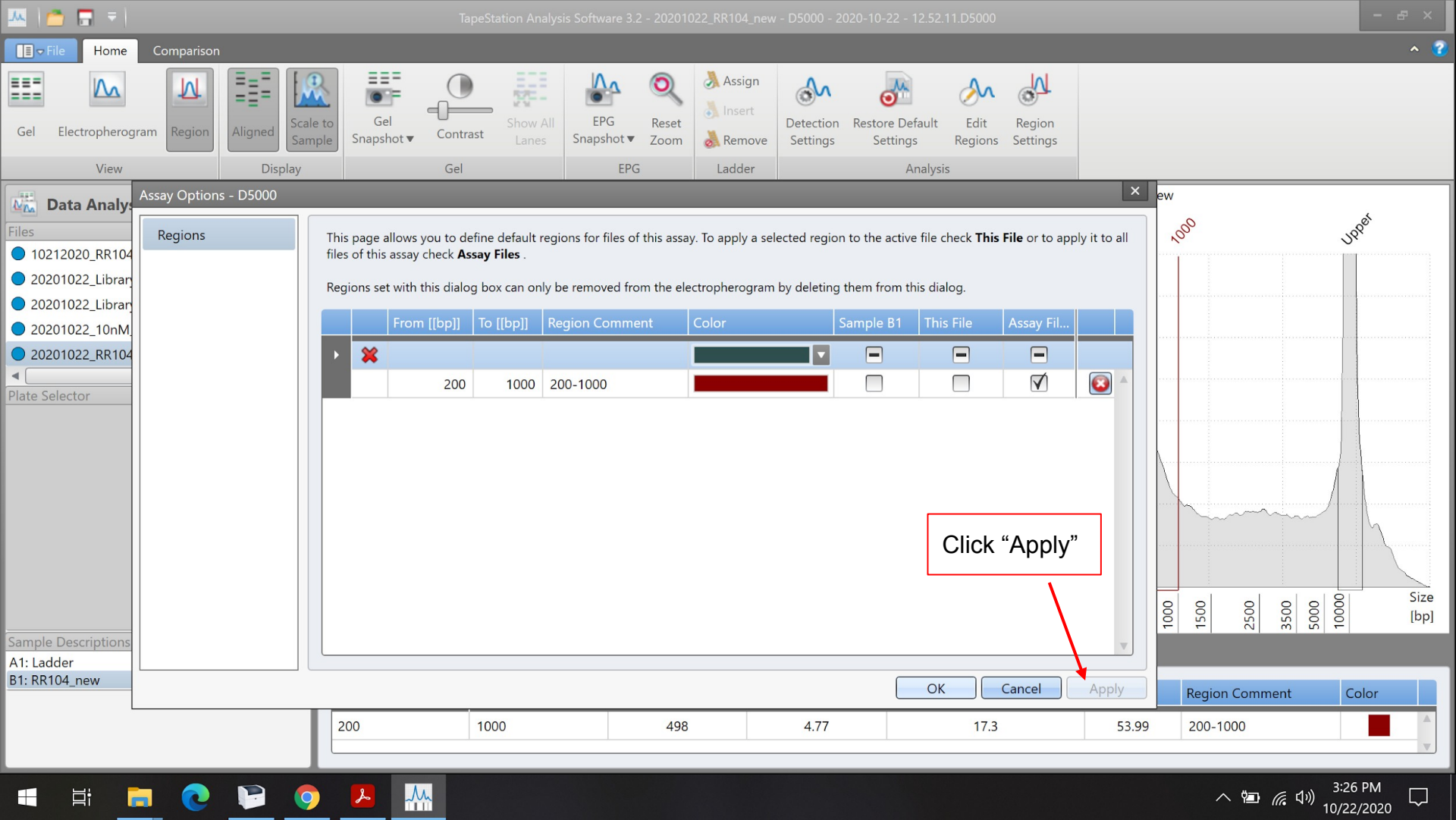
Region Table | Sample Table

From [bp]	To [bp]	Average Size [bp]	Conc. [ng/µl]	Region Molarity [nmol/l]	% of Total	Region Comment	Color

3:22 PM
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TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000
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Restore Default Settings
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Ladder
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Data Analysis
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10212020_RR104
20201022_Library
20201022_Library
20201022_10nM_
20201022_RR104
Plate Selector
Sample Descriptions
A1: Ladder
B1: RR104_new
Assay Options - D5000
Regions
This page allows you to define default regions for files of this assay. To apply a selected region to the active file check This File or to apply it to all files of this assay check Assay Files .
Regions set with this dialog box can only be removed from the electropherogram by deleting them from this dialog.
From [[bp]]
To [[bp]]
Region Comment
Color
Sample B1
This File
Assay Fil...
Click here to add a new row
Click on this line to add the peak range of interest
OK
Cancel
Apply
Upper
1000
1500
2500
3500
5000
10000
Size [bp]
Region Comment
Color
3:24 PM
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TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000
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EPG Snapshot
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Insert
Remove
Detection Settings
Restore Default Settings
Edit Regions
Region Settings
View
Display
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Ladder
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Data Analysis
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10212020_RR104
20201022_Library
20201022_Library
20201022_10nM
20201022_RR104
Plate Selector
Sample Descriptions
A1: Ladder
B1: RR104_new
Assay Options - D5000
Regions
This page allows you to define default regions for files of this assay. To apply a selected region to the active file check This File or to apply it to all files of this assay check Assay Files .
Regions set with this dialog box can only be removed from the electropherogram by deleting them from this dialog.
From [[bp]]
To [[bp]]
Region Comment
Color
Sample B1
This File
Assay Fil...
200
1000
200-1000
Enter the “from” and “to” range here. We commonly use the 200-1000 bp range
By selecting “assay file” this range will apply to all future assays. But you can also specify the range for a sample or file.
OK
Cancel
Apply
Upper
1000
1500
2500
3500
5000
10000
Size [bp]
Region Comment
Color
3:26 PM
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TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000
File
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10212020_RR104
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20201022_10nM_
20201022_RR104
Plate Selector
Sample Descriptions
A1: Ladder
B1: RR104_new
Assay Options - D5000
Regions
This page allows you to define default regions for files of this assay. To apply a selected region to the active file check This File or to apply it to all files of this assay check Assay Files .
Regions set with this dialog box can only be removed from the electropherogram by deleting them from this dialog.
From [[bp]]
To [[bp]]
Region Comment
Color
Sample B1
This File
Assay Fil...
200
1000
200-1000
Click “Apply”
OK
Cancel
Apply
1000
Upper
1000
1500
2500
3500
5000
10000
Size [bp]
Region Comment
Color
200
1000
498
4.77
17.3
53.99
200-1000
3:26 PM
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TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000

File | Home | Comparison

Gel | Electropherogram | Region | Aligned | Scale to Sample | Gel Snapshot | Contrast | Show All Lanes | EPG Snapshot | Reset Zoom | Assign | Insert | Remove | Detection Settings | Restore Default Settings | Edit Regions | Region Settings

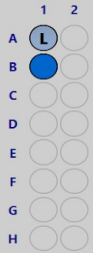
View | Display | Gel | EPG | Ladder | Analysis

Data Analysis

Files

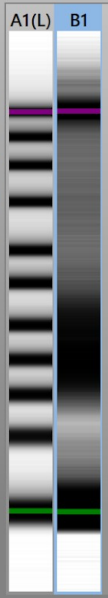
- 10212020_RR104_FS_DNAse_ARTIC - HSD5000
- 20201022_LibraryQC - D5000 - 2020-10-22 - 1
- 20201022_LibraryQC_HS - HSD5000 - 2020-10
- 20201022_10nM_PhiX_stocks - D5000 - 2020-1
- 20201022_RR104_new - D5000 - 2020-10-22 -

Plate Selector

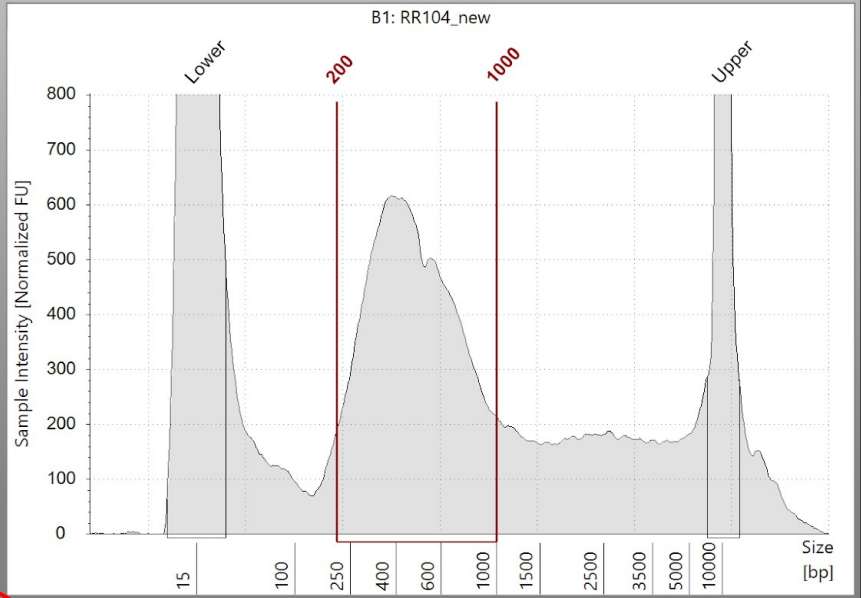


Sample Descriptions

A1: Ladder
B1: RR104_new



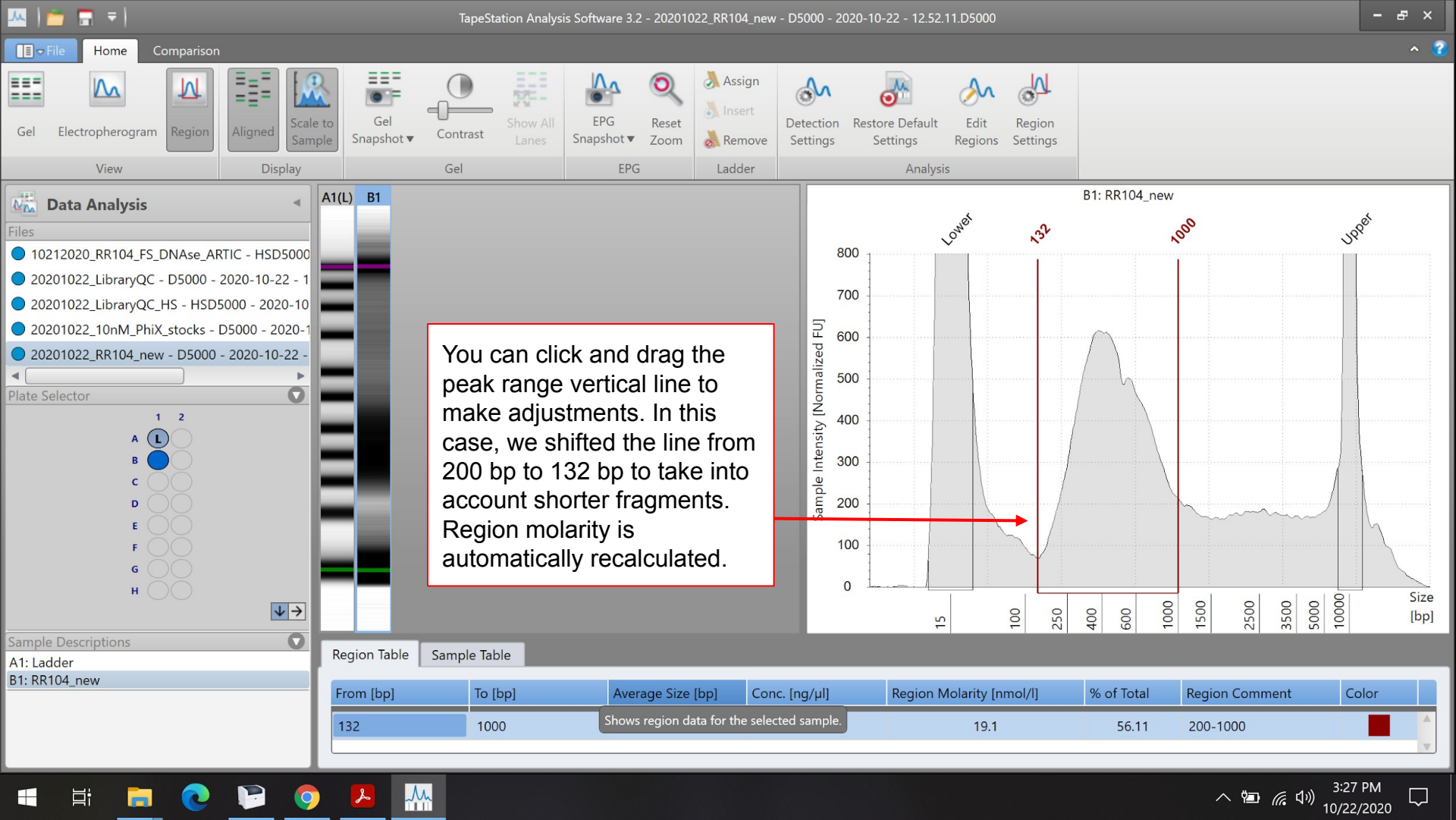
The “Region Molarity” concentration is now based on the 200-1000 bp range selected. This is the TS value that we use for our sample quantification

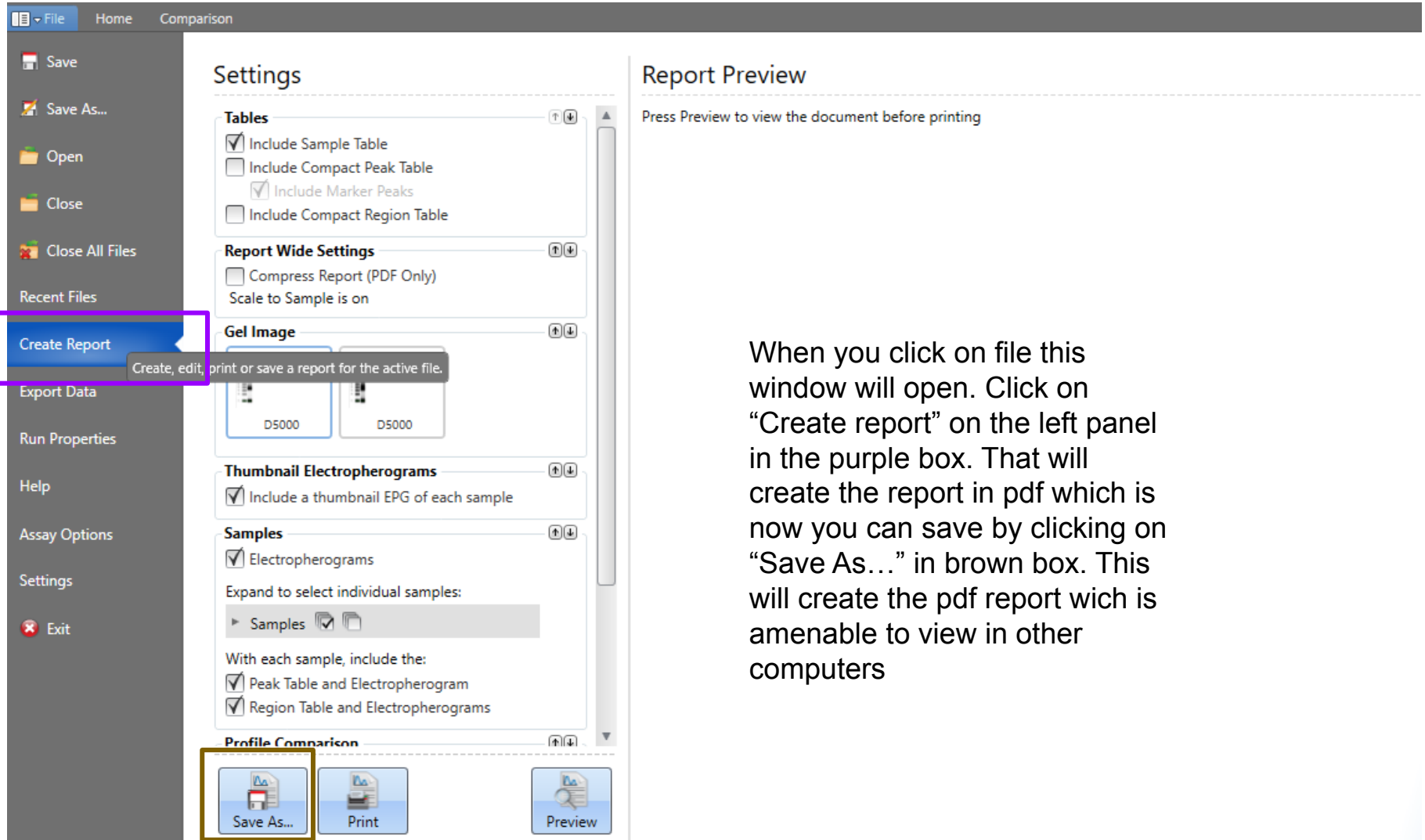


Region Table | Sample Table

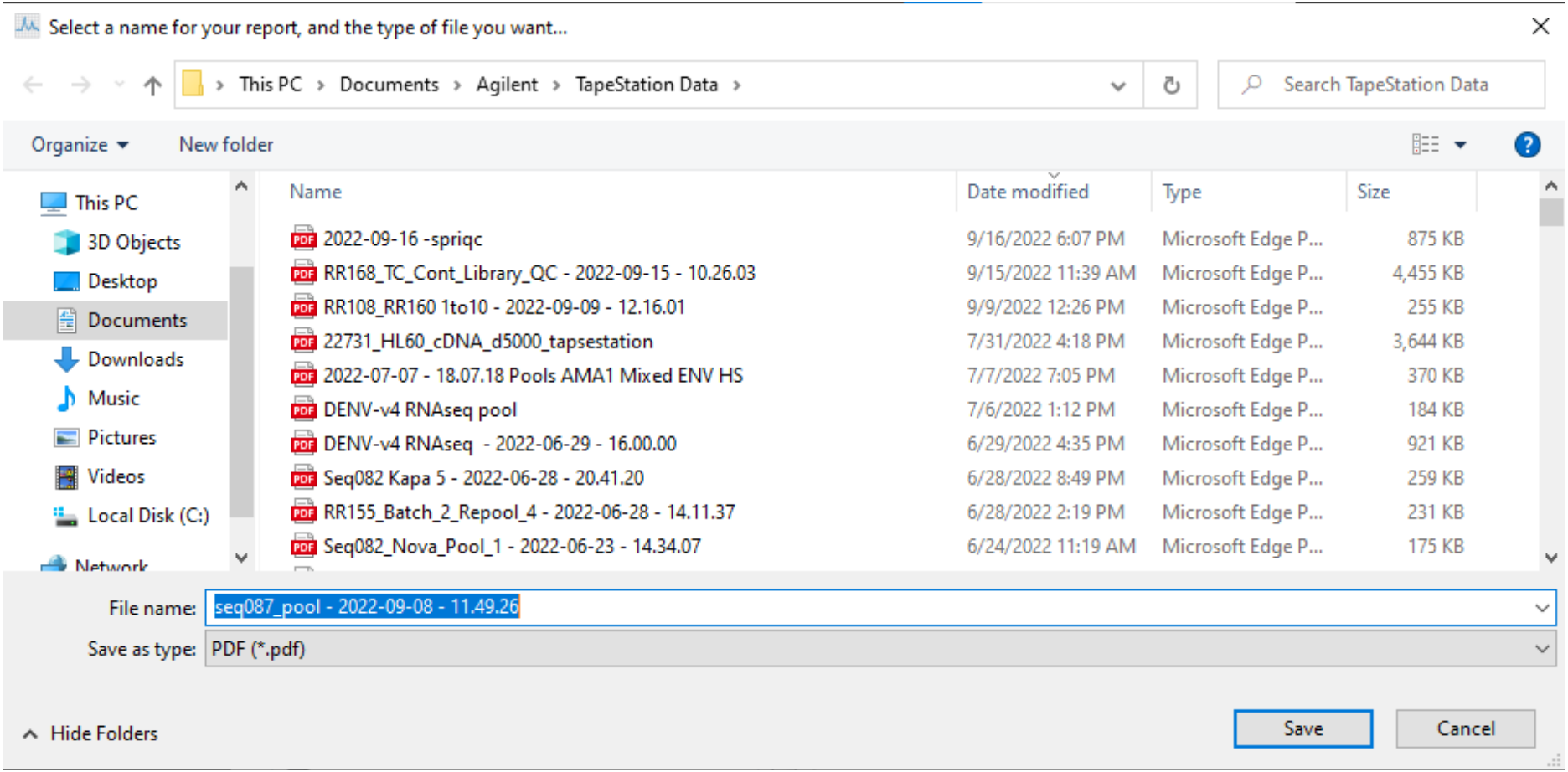
From [bp]	To [bp]	Average Size [bp]	Conc. [ng/µl]	Region Molarity [nmol/l]	% of Total	Region Comment	Color
200	1000	498	4.77	17.3	53.99	200-1000	

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TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000
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10212020_RR104_FS_DNAse_ARTIC - HSD5000
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20201022_LibraryQC_HS - HSD5000 - 2020-10
20201022_10nM_PhiX_stocks - D5000 - 2020-1
20201022_RR104_new - D5000 - 2020-10-22 -
Plate Selector
Sample Descriptions
A1: Ladder
B1: RR104_new
A1(L)
B1
You can click and drag the peak range vertical line to make adjustments. In this case, we shifted the line from 200 bp to 132 bp to take into account shorter fragments. Region molarity is automatically recalculated.
B1: RR104_new
Lower
132
1000
Upper
Sample Intensity [Normalized FU]
Size [bp]
Region Table
Sample Table
From [bp]
To [bp]
Average Size [bp]
Conc. [ng/µl]
Region Molarity [nmol/l]
% of Total
Region Comment
Color
132
1000
Shows region data for the selected sample.
19.1
56.11
200-1000
3:27 PM
10/22/2020



When you click on file this window will open. Click on “Create report” on the left panel in the purple box. That will create the report in pdf which is now you can save by clicking on “Save As…” in brown box. This will create the pdf report wich is amenable to view in other computers



Save and now your report is ready to share as a pdf!

Next: 06-I_Illumina sequencing